**State-wise Analysis of Aadhaar Biometric Update Behaviour**

## 1. About

This project presents a data-driven analytical study conducted as part of the **UIDAI Data Hackathon 2026**, focusing on understanding and characterizing **Aadhaar biometric update behaviour across Indian states**.

Using the official Aadhaar biometric datasets provided by UIDAI, the analysis applies **statistical aggregation, feature engineering, anomaly detection, and unsupervised clustering** techniques to uncover meaningful patterns in biometric update activity. The objective is to support **operational monitoring, anomaly identification, and evidence-based decision-making** at a national scale.

All analysis has been performed strictly on the datasets supplied for the hackathon, adhering to UIDAI data usage guidelines.

## 2. Problem Statement

UIDAI handles biometric updates at massive scale across diverse geographic regions. Due to differences in population size, infrastructure, administrative processes, and operational conditions, biometric update activity may exhibit:

- Significant variation across states
- Irregular spikes or drops over time
- States that behave structurally differently from others

The core problem addressed in this project is:

**How can we systematically group Indian states based on biometric update behaviour to identify stable regions, high-activity regions, and anomaly-prone regions using data-driven techniques?**

## 3. Datasets Used

The analysis exclusively uses **Aadhaar biometric update datasets** released by UIDAI for the hackathon.

**Key Data Fields**

- **Geographical identifiers:** State, District, Pincode
- **Temporal attributes:** Date, Month
- **Operational metrics:**
    - bio_age_5_17
    - bio_age_17_

A derived metric representing **total biometric updates** was created as:

metric = bio_age_5_17 + bio_age_17_

| | date | state | district | pincode | bio_age_5_17 | bio_age_17_ | new_date | month | state_clean | metric |
|---|---|---|---|---|---|---|---|---|---|---|
| **4687** | 01-03-2025 | Andaman & Nicobar Islands | Andamans | 744101 | 16 | 193 | 20250301 | 03 | Andaman and Nicobar Islands | 209 |
| **38211** | 01-04-2025 | Andaman & Nicobar Islands | Andamans | 744101 | 17 | 167 | 20250401 | 04 | Andaman and Nicobar Islands | 184 |
| **59019** | 01-05-2025 | Andaman & Nicobar Islands | Andamans | 744101 | 12 | 128 | 20250501 | 05 | Andaman and Nicobar Islands | 140 |
| **67056** | 01-06-2025 | Andaman & Nicobar Islands | Andamans | 744101 | 11 | 129 | 20250601 | 06 | Andaman and Nicobar Islands | 140 |
| **02293** | 01-07-2025 | Andaman & Nicobar Islands | Andamans | 744101 | 20 | 239 | 20250701 | 07 | Andaman and Nicobar Islands | 259 |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **15309** | 12-11- | odisha | Gajapati | 761208 | 0 | 1 | 20251112 | 11 | Odisha | 1 |

**4. Data Cleaning and Preprocessing**

The following preprocessing steps were applied:

- Standardization of **state names** to resolve spelling and naming inconsistencies
- Removal of invalid or inconsistent records
- Conversion of dates into standardized formats
- Aggregation of data at **state–date** and **state-level** granularity

These steps ensured clean, reliable inputs for further analysis.

**5. Feature Engineering**

To characterize biometric behaviour at the **state level**, multiple statistical features were derived:

- **Total biometric updates**
- **Average daily updates**
- **Median daily updates**
- **Standard deviation (volatility)**
- **Maximum daily updates**
- **Zero-activity ratio** (fraction of days with zero updates)

These features jointly capture **volume, consistency, variability, and inactivity patterns** for each state.

## 6. Methodology

The methodology adopted in this project follows a structured data analytics pipeline, combining data preprocessing, feature engineering, anomaly detection, dimensionality reduction, and clustering to analyze Aadhaar biometric update behaviour at the state level.

### 6.1 Data Preprocessing and Standardization

The raw Aadhaar biometric datasets were first cleaned and standardized to ensure analytical consistency:

- State names were normalized using rule-based text cleaning and mapping to resolve spelling variations and historical naming inconsistencies.
- Date fields were converted into standardized formats to enable temporal aggregation.
- A unified dataset was created by merging multiple biometric files into a single analytical table.
- Invalid or inconsistent records were filtered out.

These steps ensured that subsequent analysis was performed on reliable and consistent data.

### 6.2 Metric Construction

To represent biometric update activity, a consolidated biometric update metric was created for each record:

$$\text{Metric} = \text{bio_age_5_17} + \text{bio_age_17_}$$

This metric captures the total number of biometric updates performed across age groups and serves as the primary variable for behavioural analysis.

### 6.3 Temporal Aggregation

To analyze behavioural patterns rather than individual transactions, the data was aggregated at multiple levels:

- State–Date level aggregation to compute daily biometric activity per state.
- State-level aggregation to summarize long-term behaviour across the full observation period.

This transformation enabled the study of sustained trends rather than isolated events.

### 6.4 Feature Engineering

A comprehensive set of features was engineered to capture multiple dimensions of biometric update behaviour:

#### 6.4.1 Volume-Based Features

- Total biometric updates: Cumulative updates across all dates
- Average daily updates: Mean daily activity
- Median daily updates: Central tendency resistant to outliers
- Maximum daily updates: Peak activity levels

These features describe the overall scale of biometric activity.

### 6.4.2 Variability and Stability Features

- Standard deviation of daily updates: Measures volatility and fluctuation
- Zero-activity ratio: Proportion of days with zero biometric updates

These features identify states with stable reporting versus irregular or inconsistent behaviour.

### 6.4.3 Anomaly-Oriented Features

To capture irregular patterns, multiple anomaly indicators were computed:

- Percentage change anomaly rate: Flags sudden spikes or drops greater than 100% relative to previous observations.
- Spatial anomaly ratio: Compares pincode-level activity to district-level averages to detect spatial inconsistencies.
- IQR-based anomaly rate: Identifies outliers using the Interquartile Range method within geographic groupings.

These features quantify abnormal behaviour rather than relying on raw visual inspection.

## 6.5 Feature Scaling

Since the engineered features exist on different numerical scales, standardization using z-score normalization was applied. This step ensures that no single feature dominates the clustering process due to scale differences.

## 6.6 Dimensionality Reduction

To improve interpretability and visualization:

- Principal Component Analysis (PCA) was applied to the standardized feature set.
- The first two principal components were used to project states into a two-dimensional behavioural space.

This step preserves maximum variance while enabling visual interpretation of cluster separation.

## 6.7 Clustering Technique

- K-Means clustering was applied to the scaled feature set.
- The optimal number of clusters was determined using the Elbow Method, balancing model simplicity and explanatory power.
- Each state was assigned to a cluster representing a distinct biometric update behaviour profile.

## 6.8 Cluster Interpretation

Clusters were interpreted using aggregated feature statistics to understand:

- Activity intensity
- Consistency and stability
- Volatility and anomaly prevalence

This allowed the categorization of states into meaningful behavioural groups such as stable, moderate, volatile, and high-volume regions.

**7. Cluster Interpretation and Insights**

The clustering process resulted in **four distinct groups of states**:

**Cluster 0 – *Low-Volume Stable States***

- Low biometric activity
- High consistency
- Minimal anomalies
- Typically smaller or less populous regions

**Cluster 1 – *Moderate-Volume Consistent States***

- Steady and regular biometric updates
- Moderate volume
- Operationally stable behaviour

**Cluster 2 – *Volatile / Anomaly-Prone States***

- High variability in biometric updates
- Sudden spikes or drops
- Potential operational irregularities

**Cluster 3 – *High-Volume Outlier States***

- Exceptionally high biometric update volumes
- Structurally distinct from other states
- Often large and highly populous regions

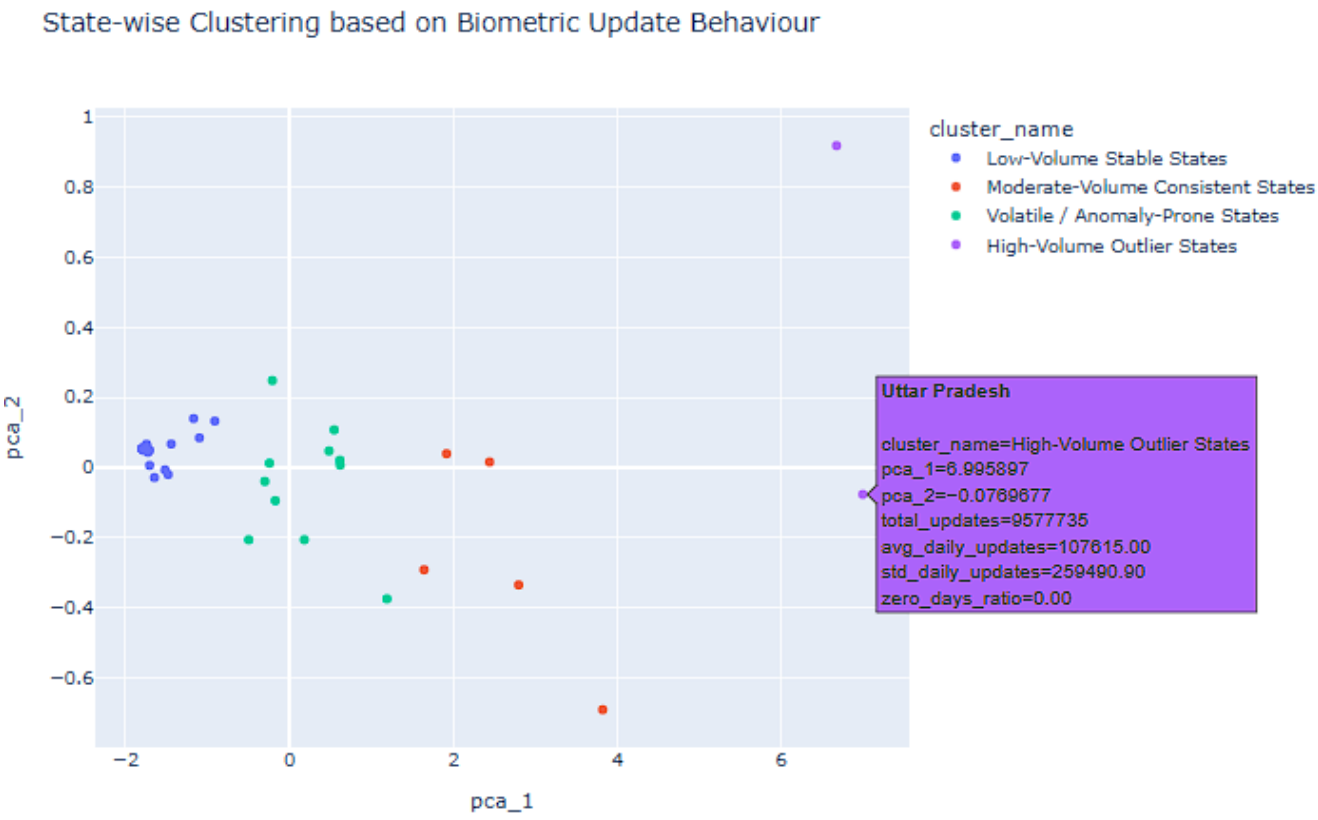
This grouping enables **quick identification of operational patterns** and helps distinguish normal structural differences from potential anomalies.

**8. Visualisation**

An interactive **PCA-based scatter plot** was developed to visualize state clusters:

- Each point represents a state
- Color indicates cluster membership
- Hover interaction displays:
    - State name
    - Update volume statistics
    - Variability indicators

These visualisations enhance interpretability for **non-technical stakeholders** and decision-makers.




**9. Key Outcomes**

- Clear segmentation of Indian states based on biometric update behaviour
- Identification of structurally high-volume and anomaly-prone regions
- Scalable framework suitable for continuous monitoring
- Interpretable results aligned with operational decision-making needs

**10. Conclusion**

```
state_features.groupby('cluster_name')['state_clean'].apply(list)
```

```
cluster_name
High-Volume Outlier States                          [Maharashtra, Uttar Pradesh]
Low-Volume Stable States            [Andaman and Nicobar Islands, Arunachal Prades...
Moderate-Volume Consistent States   [Andhra Pradesh, Bihar, Madhya Pradesh, Rajast...
Volatile / Anomaly-Prone States     [Chhattisgarh, Delhi, Gujarat, Haryana, Jharkh...
Name: state_clean, dtype: object
```

The state-wise clustering of Aadhaar biometric update data successfully revealed **distinct behavioural patterns across Indian states**, enabling meaningful segmentation based on update volume, stability, and anomaly characteristics.

The analysis identified **High-Volume Outlier States**, including **Maharashtra and Uttar Pradesh**, which exhibit exceptionally high biometric update activity and dominate overall system volume. These states form structurally distinct outliers and may require focused capacity planning and continuous operational monitoring.

The **Low-Volume Stable States** cluster comprises regions such as **Andaman and Nicobar Islands, Arunachal Pradesh, and Sikkim**, which demonstrate consistently low biometric activity with minimal variability, indicating stable and predictable update behaviour.

States such as **Andhra Pradesh, Bihar, Madhya Pradesh, and Rajasthan** fall under the **Moderate-Volume Consistent States** cluster, reflecting regular and balanced biometric update patterns with moderate volume and limited anomalies.

The **Volatile / Anomaly-Prone States** cluster includes **Chhattisgarh, Delhi, Gujarat, Haryana, and Jharkhand**, which show higher variability and irregular biometric update behaviour, potentially influenced by seasonal effects, administrative differences, or reporting inconsistencies.

Overall, this clustering framework transforms large-scale biometric transaction data into **actionable, state-level behavioural insights**, enabling UIDAI to distinguish structurally high-demand regions, stable low-activity regions, and anomaly-prone regions. The methodology provides a scalable foundation for targeted monitoring, operational planning, and data-driven governance.